competishun
The Power of Real Gurus

Center of Mass, Momentum & Collisions

Quick Revision Notes · Theory & Standard Results · JEE Main / Advanced · NEET

Centre of Mass — Definition & Symmetry

Definition

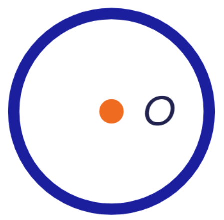
The **centre of mass** is the mass-weighted average position of a system — it moves as if the whole mass were concentrated there and all external forces acted at that point.

$$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots}{m_1 + m_2 + \cdots} = \frac{\sum m_i\,\vec{r}_i}{\sum m_i}$$

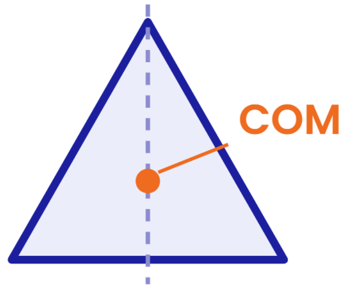
$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}, \quad y_{cm} = \frac{\sum m_i y_i}{\sum m_i}, \quad z_{cm} = \frac{\sum m_i z_i}{\sum m_i}$$

Two particles: COM divides the line joining them in the inverse ratio of masses — $m_1 r_1 = m_2 r_2$.

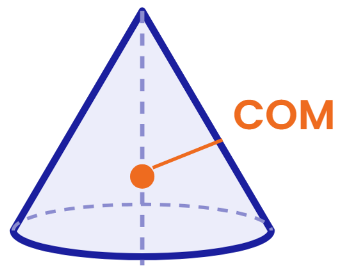
Symmetry rules — spot the COM instantly



Point symmetry — ring, disc, sphere: COM at centre O



Line symmetry — COM lies on the symmetry axis

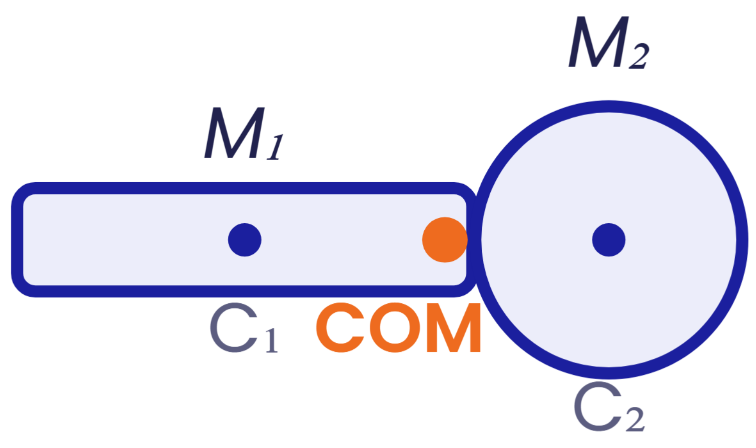


Plane symmetry — COM lies in the symmetry plane (on the axis)

COM need not lie inside the material of the body — e.g. the centre of a ring or an L-shaped plate.

Methods to Find the COM

Composite bodies



$$\vec{R}_{cm} = \frac{M_1\vec{R}_1 + M_2\vec{R}_2}{M_1 + M_2}$$

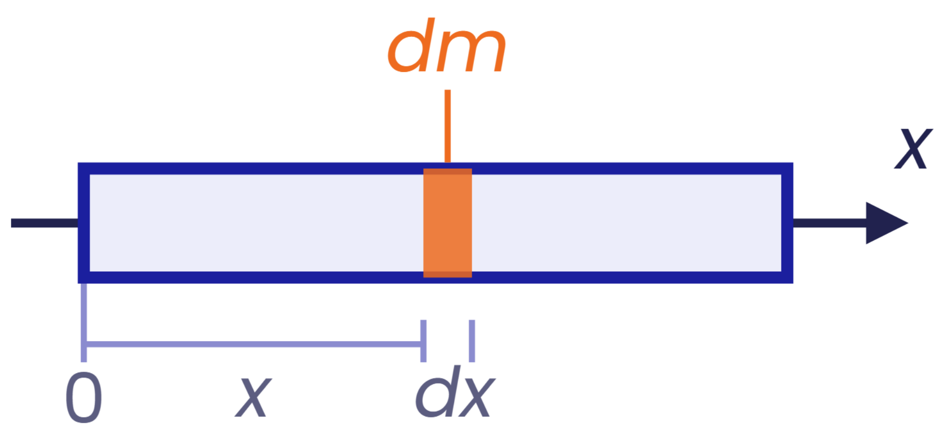
Replace each part by a point mass at its own COM $(\vec{R}_1, \vec{R}_2)$, then combine.

Cut or hole — negative mass

$$\vec{R}_{cm} = \frac{M\,\vec{r}_O - m\,\vec{r}_{hole}}{M - m}$$

Treat the cavity as mass -m at the hole centre; M = full uncut body. COM shifts away from the hole.

Continuous bodies — integration



$$\vec{r}_{cm} = \frac{1}{M}\int \vec{r}\,dm\;; \quad x_{cm} = \frac{1}{M}\int x\,dm\,, \; y_{cm} = \frac{1}{M}\int y\,dm$$

Element dm from density —

$dm = \lambda\,dx$: wire

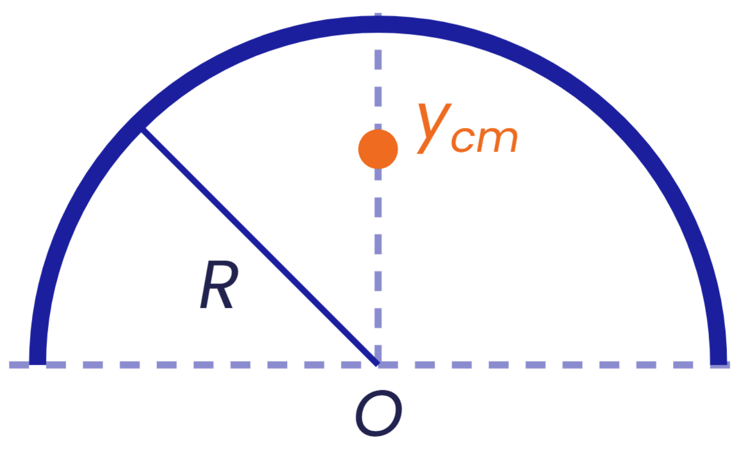
$dm = \sigma\,dA$: lamina

$dm = \rho\,dV$: solid

competishun
The Power of Real Gurus

Standard Results — 2D Bodies

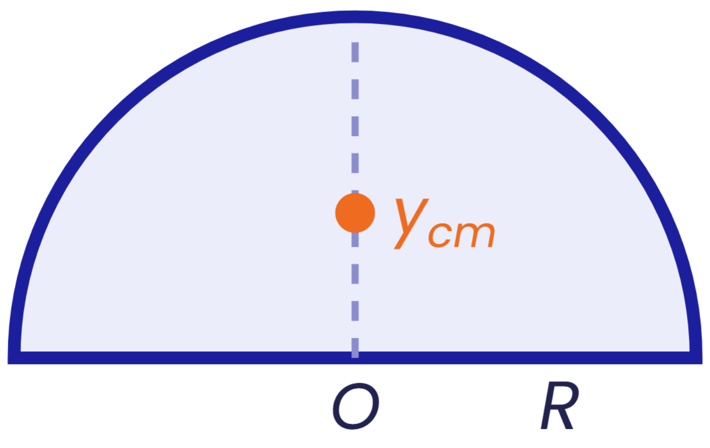
Semicircular wire



$$y_{cm} = \frac{2R}{\pi} \approx 0.64\,R$$

above the centre O, on the axis

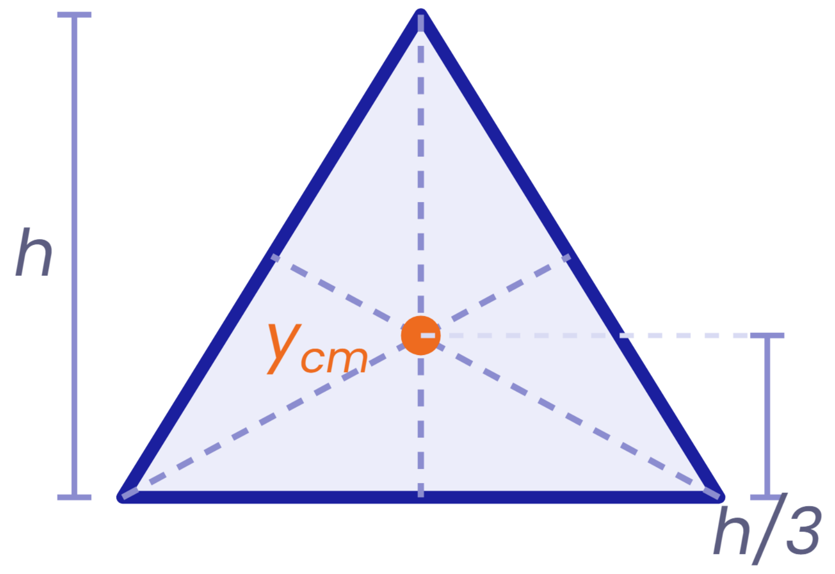
Semicircular lamina



$$y_{cm} = \frac{4R}{3\pi} \approx 0.42\,R$$

above the centre O, on the axis

Triangular lamina

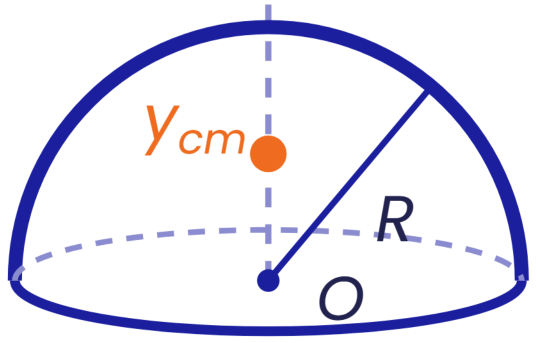


$$y_{cm} = \frac{h}{3} \text{ from base}$$

centroid — intersection of medians

Standard Results — 3D Bodies

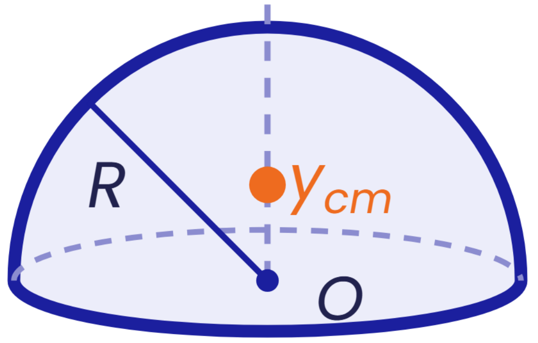
Hemispherical shell



$$y_{cm} = \frac{R}{2}$$

above centre O

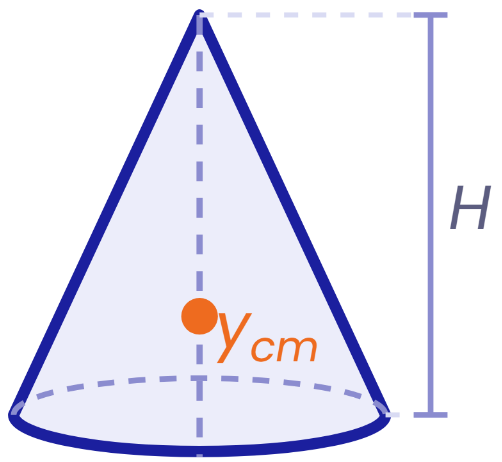
Solid hemisphere



$$y_{cm} = \frac{3R}{8}$$

above centre of base

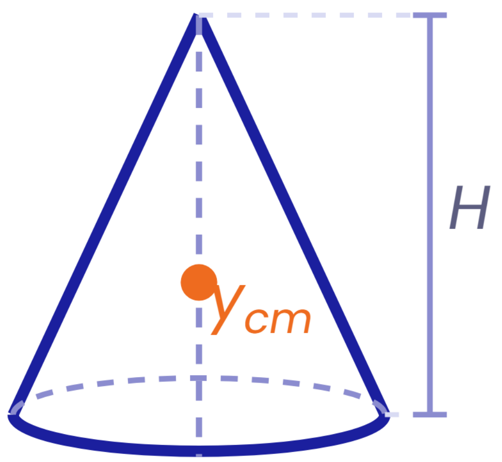
Solid cone



$\frac{H}{4}$ from base

= 3H/4 from the tip

Hollow cone



$\frac{H}{3}$ from base

= 2H/3 from the tip

Motion of the Centre of Mass

System kinematics

$$\vec{v}_{cm} = \frac{\sum m_i \vec{v}_i}{M}, \quad \vec{P}_{sys} = M\,\vec{v}_{cm}$$

$$\vec{a}_{cm} = \frac{\sum m_i \vec{a}_i}{M}$$

KEY $$\vec{F}_{ext} = M\,\vec{a}_{cm}$$

Newton's second law for a system — internal forces cancel in action–reaction pairs, so only external forces accelerate the COM.

Conservation of linear momentum

$$\vec{F}_{ext} = 0 \;\Rightarrow\; \vec{P}_{sys} = M\vec{v}_{cm} = \text{ constant}$$

Holds **componentwise**: $F_x = 0 \Rightarrow P_x = \text{ constant}$ – similarly for y and z.

Conservation of COM position (CCOM)

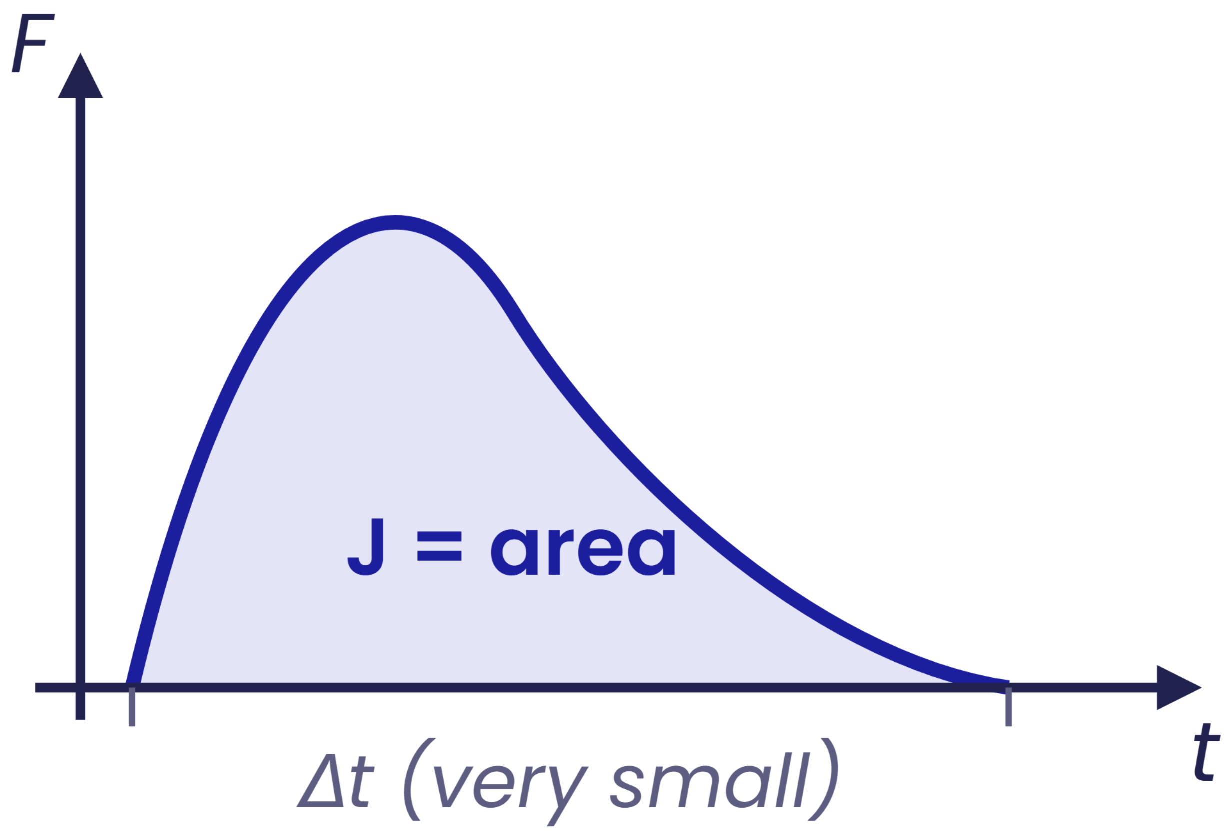
$$\vec{F}_{ext} = 0 \text{ and } \vec{v}_{cm} = 0 \;\Rightarrow\; \vec{r}_{cm} \text{ stays fixed}$$

$$\sum m_i\,\Delta\vec{r}_i = 0 \quad \text{e.g.} \quad m_1\Delta x_1 + m_2\Delta x_2 = 0$$

Displacements measured w.r.t. the ground — use when bodies exchange positions on a smooth surface.

Impulse & Impulsive Forces

Impulse = area under the F–t graph



$$\vec{J} = \int \vec{F}\,dt$$

KEY $\vec{J} = \Delta\vec{P} = \vec{P}_f - \vec{P}_i$

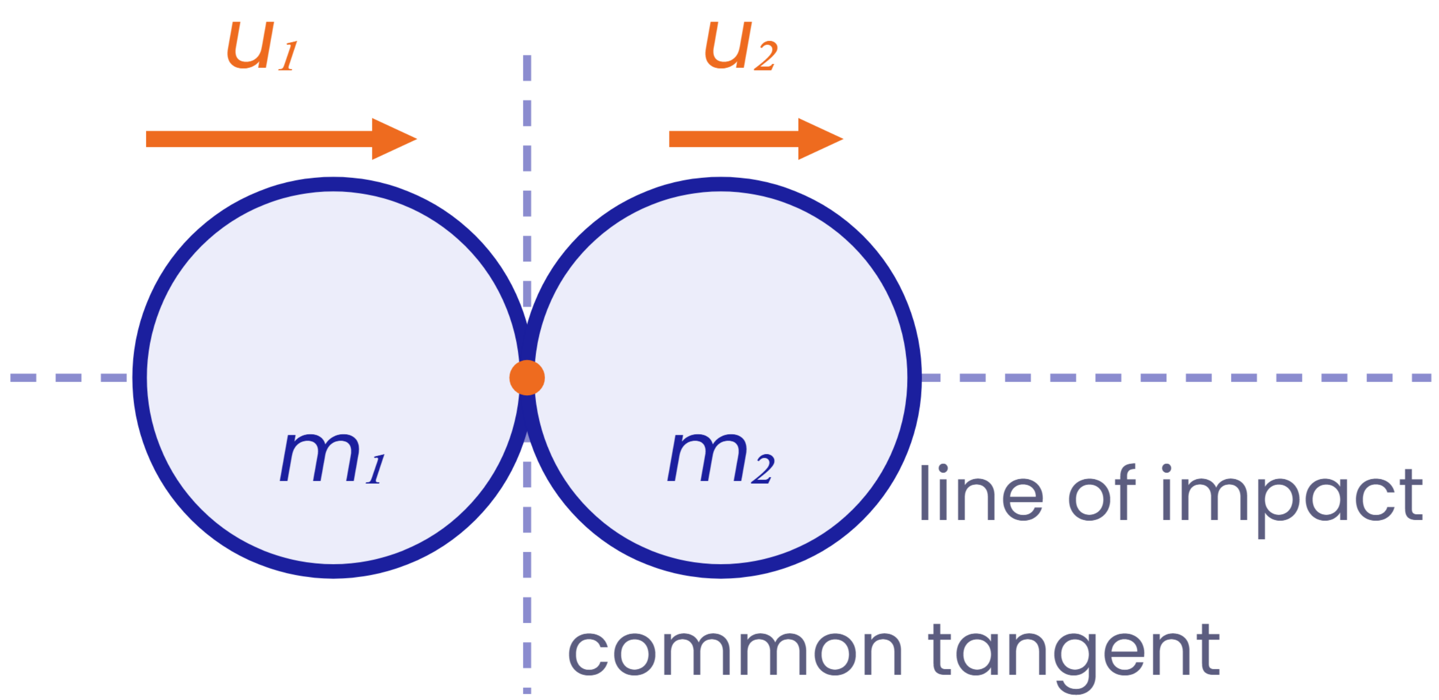
Impulsive Forces & the FID

- An **impulsive force** is very large and lasts a very short Δt — yet its impulse $\vec{J} = \vec{F}\,\Delta t$ stays finite.
- **FID (force–impulse diagram):** for the instant of impact, mark the impulse J on each body instead of forces. Action–reaction impulses are equal and opposite.
- Apply $\vec{J} = \Delta\vec{P}$ to each body along the impulse directions.

Note: during sudden jerks and impacts, neglect the impulse of finite forces — weight Mg, spring force — since $J = F\,\Delta t \to 0$ as $\Delta t \to 0$.

Collisions — Line of Impact & Restitution

Line of impact



$$v_{app} = u_1 - u_2$$

$$v_{sep} = v_2 - v_1$$

Line of impact = common normal at contact — for smooth spheres it passes through both centres. Approach & separation are measured **along this line**.

Coefficient of restitution

KEY $$e = \frac{v_{sep}}{v_{app}} = \frac{v_2 - v_1}{u_1 - u_2}$$

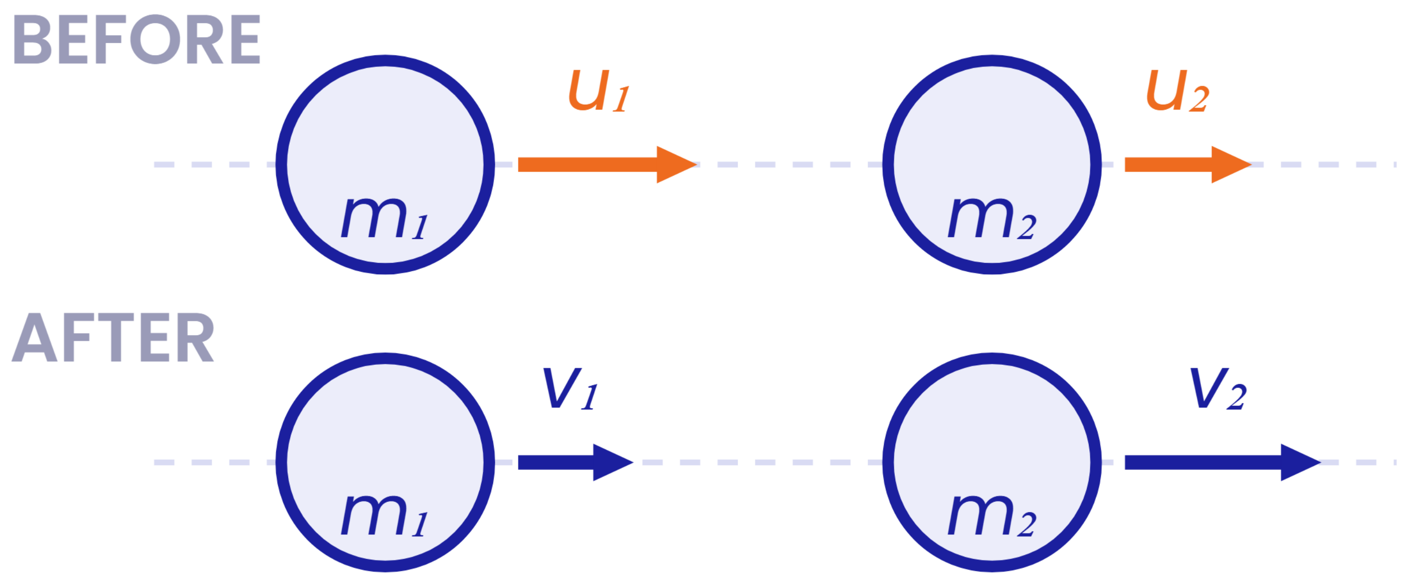
$e = 1$	$0 < e < 1$	$e = 0$
elastic — KE conserved	inelastic — KE lost	perfectly inelastic — maximum KE loss

Elastic collision between **smooth rigid spheres** $\Rightarrow e = 1$ along the line of impact.

By line of impact	head-on (1-D) / oblique (2-D)
By energy	elastic / inelastic / perfectly inelastic
By constraint	free / constraint collisions

Head-On Collision — Results & Special Cases

Setup — all along the line of impact



Two equations, two unknowns

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$v_2 - v_1 = e\,(u_1 - u_2)$$

Conserve momentum, apply restitution — solve for v_1, v_2:

KEY $$v_1 = \frac{(m_1 - em_2)\,u_1 + (1+e)\,m_2u_2}{m_1 + m_2} \qquad v_2 = \frac{(m_2 - em_1)\,u_2 + (1+e)\,m_1u_1}{m_1 + m_2}$$

Equal masses · elastic — $e = 1$

$$v_1 = u_2\,, \quad v_2 = u_1$$

Velocities are exchanged.

$m_1 \gg m_2$ · elastic

$$v_1 \approx u_1\,, \quad v_2 \approx 2u_1 - u_2$$

Heavy body moves on unchanged.

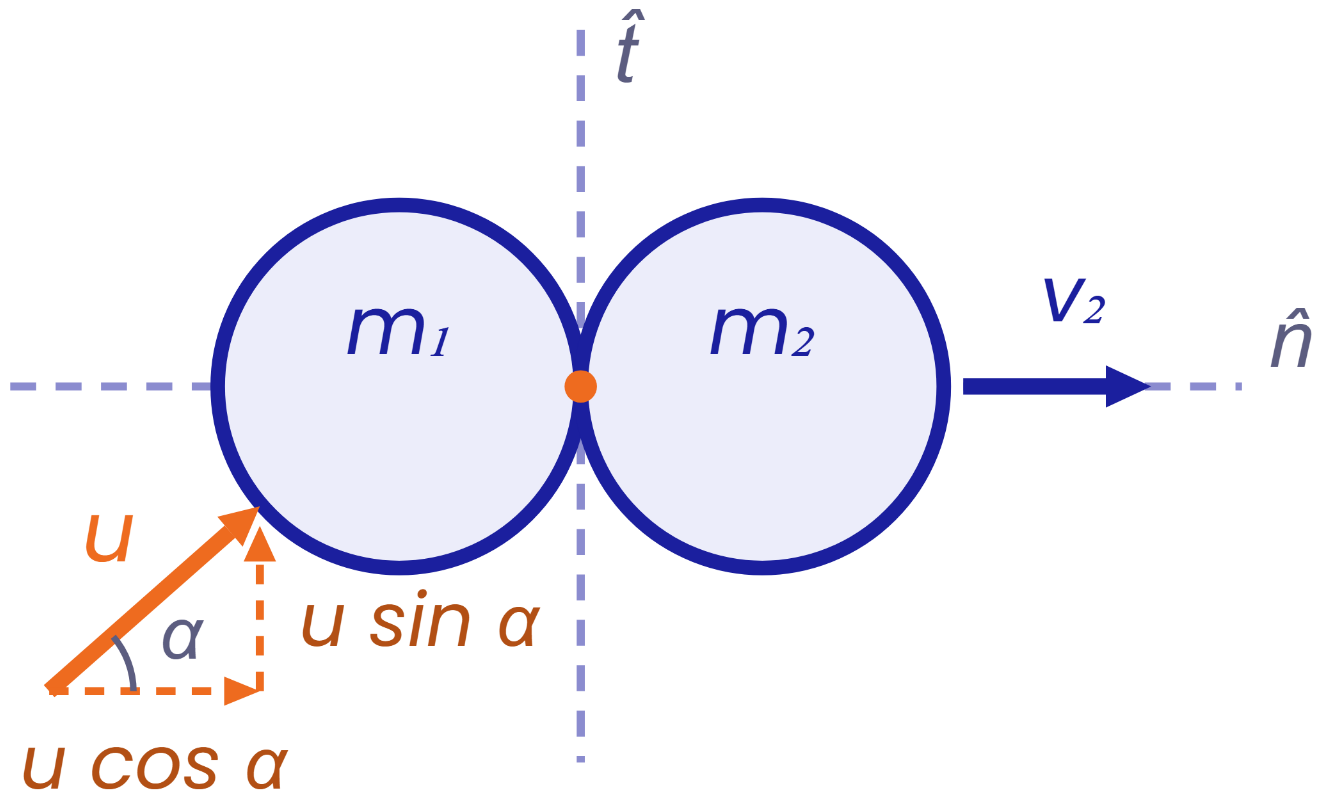
Perfectly inelastic — $e = 0$

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

Bodies move together after impact.

Oblique & Constraint Collisions

Free oblique collisions



1. Resolve velocities along n̂ (line of impact) and t̂ (common tangent).
2. Smooth bodies → no tangential impulse → each body keeps its v_t **unchanged**.
3. Along n̂: conserve momentum + apply e — exactly like a head-on collision.

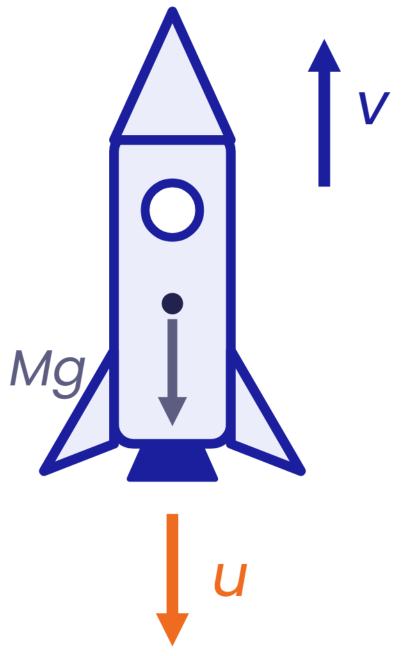
Constraint collisions

One body is not free to move arbitrarily — a hinge, rail or surface supplies an **external impulse** during the impact.

1. Conserve momentum **only along directions with no external impulse**.
2. Apply the restitution equation e along the line of impact, as usual.
3. Where an external impulse enters, write the **linear impulse–momentum** equation (LIM) $\vec{J} = \Delta\vec{P}$ for each body.

Spot the constraint: ball hitting a hinged rod, a block on a rail or a fixed wedge — never blindly conserve total momentum in every direction.

Variable-Mass Systems



exhaust · burn rate α

Rocket propulsion — thrust

$$M\frac{d\vec{v}}{dt} = \vec{F}_{ext} + \vec{v}_{rel}\frac{dm}{dt}$$

thrust force $F_t = v_{rel}\, dm/dt$

ROCKET $$M\frac{dv}{dt} = u\alpha - Mg$$

$\alpha = |dm/dt|$ = burn rate · u = exhaust speed relative to the rocket